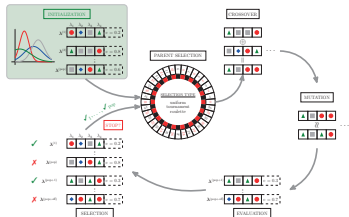
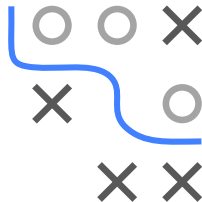


Optimization in Machine Learning

Evolutionary Algorithms Introduction

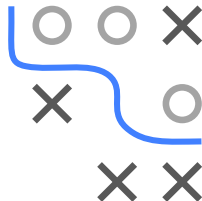


Learning goals

- Evolutionary algorithms
- Encoding
- Parent selection, variation, survival selection

NOTATION AND TERMINOLOGY

- A chromosome is a set of parameters which encodes a proposed solution to the problem that the genetic algorithm is trying to solve. The chromosome is often represented as a binary string, although a wide variety of other data structures are also used.
- The set of all solutions is known as the population.



Symbols	EA Terminology
solution candidate $\mathbf{x} \in \mathcal{S}$	chromosome of an individual
x_j	j -th gene of chromosome
set of candidates P with $\mu = P $	population and size
λ	number of generated offsprings
$f : \mathcal{S} \rightarrow \mathbb{R}$	fitness function

Note: Unintuitively, we are minimizing fitness because we always minimize f by convention.

ENCODING

Encoding of chromosomes is the first step of solving a problem with EAs. Technically: Mapping from **genotype** to **phenotype**. Encoding depends on the problem, and eventually decides performance of problem solving. **Encoding methods:**

- Binary encoding: Strings of 0s and 1s
- Real value encoding: Real values

Genotype:



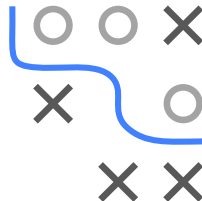
Phenotype:



Binary encoding

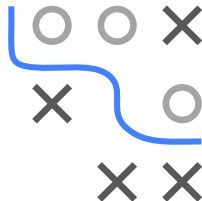


Real value encoding



STEP 1: INITIALIZE POPULATION

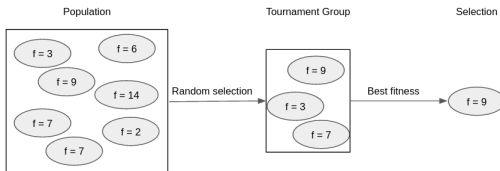
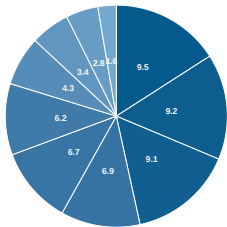
- Evolutionary algorithms start with generating initial population $P = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(\mu)}\}$.
- Usually: Initialize uniformly at random.
- Introducing prior knowledge possible.
- Population is evaluated: objective function is computed for each initial individual.
- Initialization influences quality of solution, so many EAs employ *restarts* with new randomly generated initial populations.



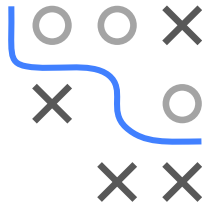
STEP 2: PARENT SELECTION

Choose a number of λ parents pairs creating λ offsprings.

- **Neutral selection:** Draw parents uniformly at random.
- **Fitness-proportional / Roulette wheel selection:** Draw individuals with probability proportional to their fitness.
- **Tournament selection:** Randomly select k individuals for a "tournament group" and pick the best one (according to fitness value).



Left: Fitness-proportional selection. Fitness values of $\mu = 10$ individuals are converted into probabilities. **Right:** Tournament selection.

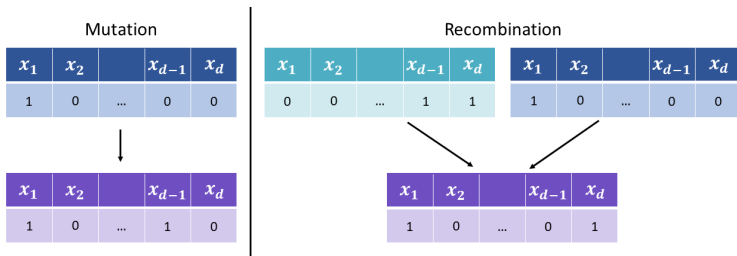
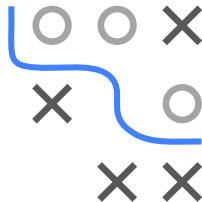


STEP 3: VARIATION

New individuals (offsprings) are generated from parents.

- Recombination/Crossover: Combine two parents into offspring.
- Mutation: Modify the offspring locally.

Sometimes only one of both operations is performed.



Note: Particular operation depends on encoding. Examples for binary and numeric encodings follow later.

EVOLUTIONARY ALGORITHMS

- **Advantages**

- Simple but enough to solve complex problems
- All parameter types possible in general
- Highly parallelizable
- Flexible through different variation operations

- **Disadvantages**

- Little mathematical rigor (for realistic, complex EAs)
- Hard to find balance between exploration and exploitation
- Quite some parameters, hard to determine them
- Customization necessary for complex problems
- Not suitable for expensive problems like HPO as large number of function evaluations necessary

